Midterm Review

Computer Organization

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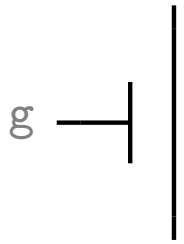
For each topic, we will **summarize** what you need to know, and then answer a few questions.

1. transistors
2. Boolean algebra
3. combinational design
4. logical devices
5. binary arithmetic
6. arithmetic circuits
7. flip-flops
8. state machines



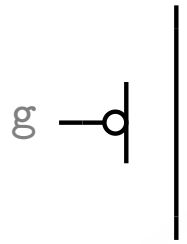
transistors

nMOS



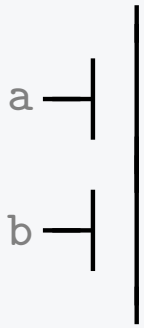
conducts when the gate is 1

pMOS



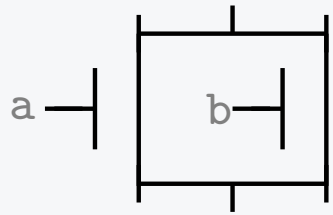
conducts when the gate is 0

in series



AND: both conduct

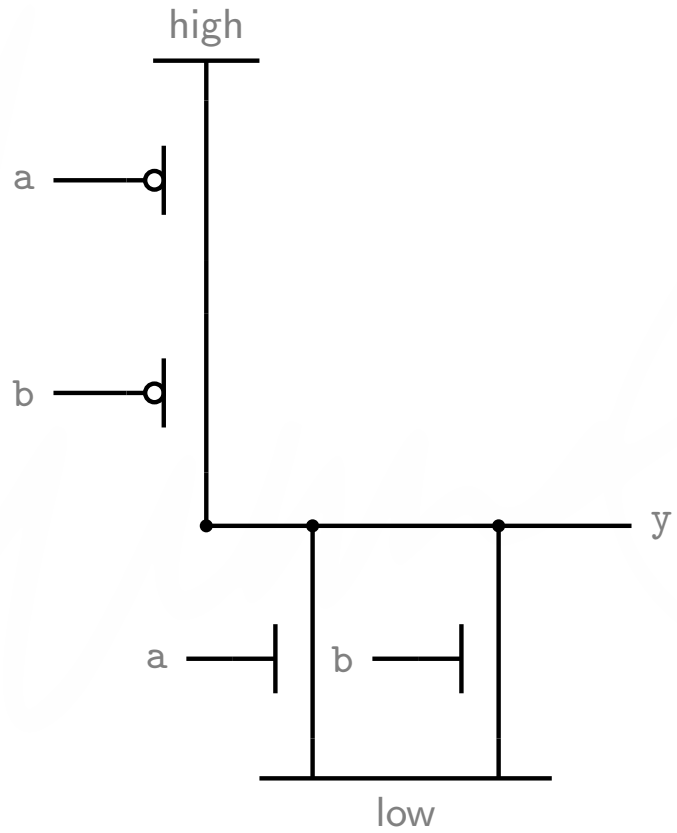
in parallel



OR: either conducts

a gate connects its output to high through one network and to low through the other

question 1

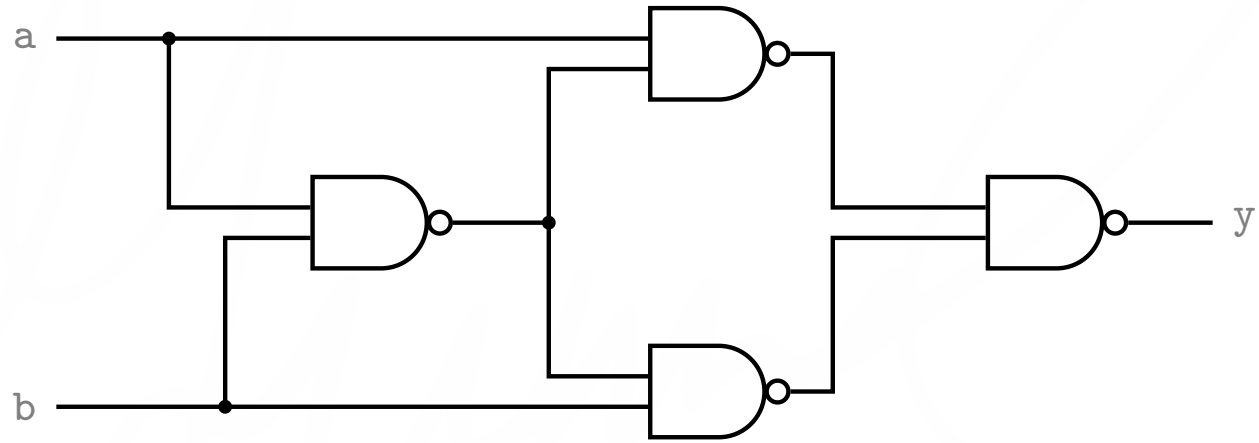


0 responses

Which gate does this transistor circuit implement?

A	AND	0%	0 votes
B	OR	0%	0 votes
C	NAND	0%	0 votes
✓	NOR	0%	0 votes

question 2



0 responses

Which gate do these four `NAND` gates build?

A	AND	0%	0 votes
B	OR	0%	0 votes
✓	XOR	0%	0 votes
D	XNOR	0%	0 votes

2

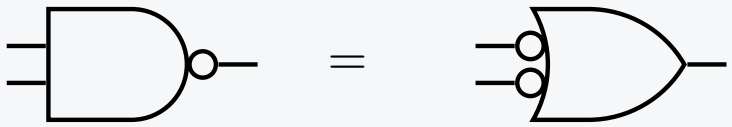
Boolean algebra

Each law comes in an AND form and an OR form.

identity	$a \cdot 1 = a$	$a + 0 = a$
null	$a \cdot 0 = 0$	$a + 1 = 1$
idempotent	$a \cdot a = a$	$a + a = a$
complement	$a \cdot \overline{a} = 0$	$a + \overline{a} = 1$
involution	$\overline{\overline{a}} = a$	
commutative	$ab = ba$	$a + b = b + a$
associative	$a(bc) = (ab)c$	$a + (b + c) = (a + b) + c$
distributive	$a(b + c) = ab + ac$	$a + bc = (a + b)(a + c)$
absorption	$a(a + b) = a$	$a + ab = a$
consensus	$ab + \overline{a}c + bc = ab + \overline{a}c$	
De Morgan	$\overline{a \cdot b} = \overline{a} + \overline{b}$	$\overline{a + b} = \overline{a} \cdot \overline{b}$

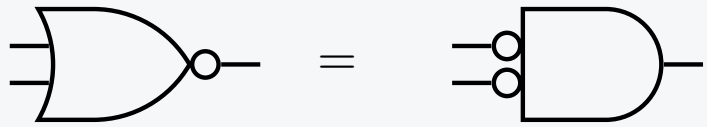
De Morgan's laws let us push a bubble through a gate: `AND` and `OR` swap, and every input and output flips.

first form



$$\overline{a \cdot b} = \overline{a} + \overline{b}$$

second form

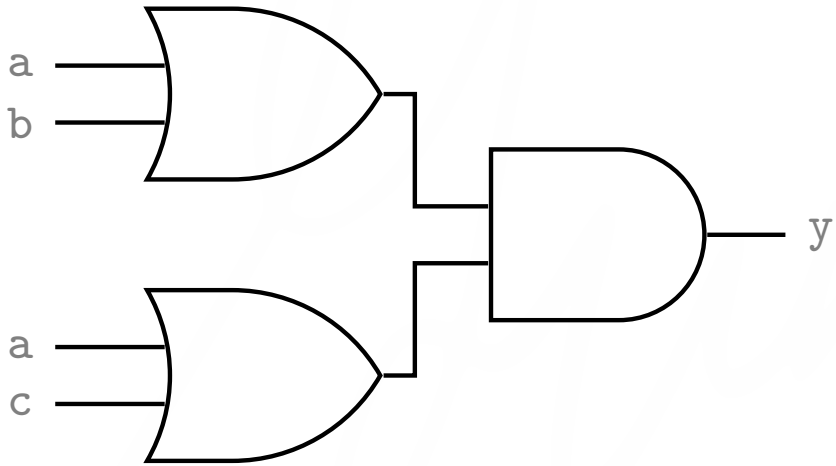


$$\overline{a + b} = \overline{a} \cdot \overline{b}$$

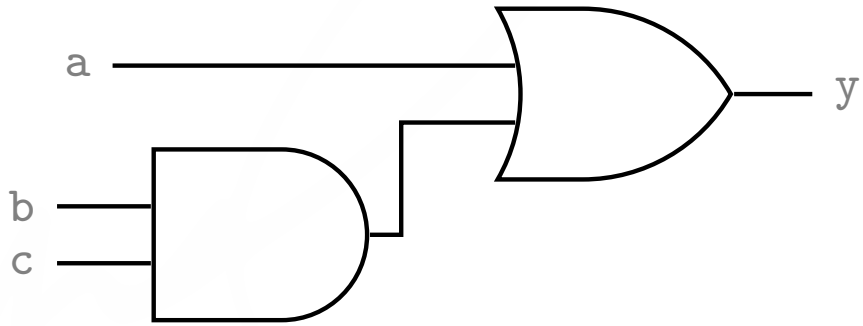
a `NAND` is an `OR` of inverted inputs; a `NOR` is an `AND` of inverted inputs

question 3

circuit A



circuit B

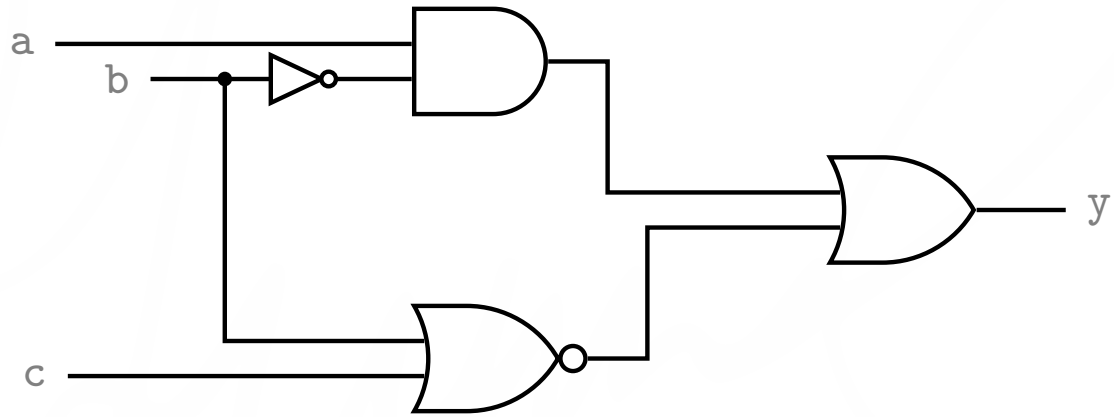


0 responses

Circuits A and B compute the same function.

✓ True	0%	0 votes
B False	0%	0 votes

question 4



0 responses

Which expression does this circuit implement?

A	$a\overline{b} + \overline{b} + \overline{c}$	0% 0 votes
	$a\overline{b} + \overline{b}\,\overline{c}$	0% 0 votes
C	$\overline{a\overline{b}} + bc$	0% 0 votes
D	$(a + \overline{b})\,\overline{b}\,\overline{c}$	0% 0 votes

0 responses

Simplify $\overline{(\overline{a}+b)\,\overline{c}}+\overline{a}c$.

A $a+c$ 0% 0 votes

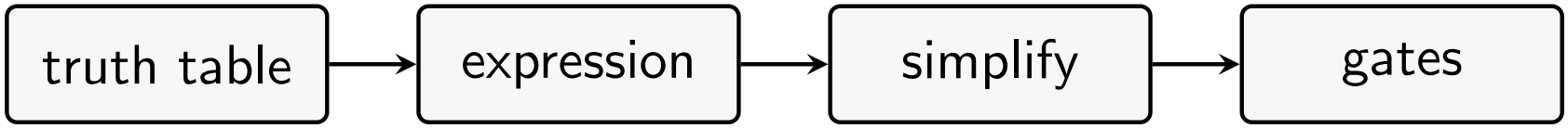
B $(\overline{a}+b)\,\overline{c}$ 0% 0 votes

✓ $a\overline{b}+c$ 0% 0 votes

D $a\overline{b}+\overline{a}c$ 0% 0 votes

3

combinational design



a	b	c	f	
0	0	0	1	m_0
0	0	1	1	m_1
0	1	0	0	M_2
0	1	1	0	M_3
1	0	0	1	m_4
1	0	1	1	m_5
1	1	0	0	M_6
1	1	1	1	m_7

minterm m_i (the shaded rows)

AND of every input: $m_5 = \mathtt{a}\,\overline{\mathtt{b}}\,\mathtt{c}$

want these rows

$$\mathtt{f} = \Sigma m(0, 1, 4, 5, 7)$$

maxterm M_i (the white rows)

OR of every input: $M_2 = \mathtt{a} + \overline{\mathtt{b}} + \mathtt{c}$

don't want these rows

$$\mathtt{f} = \Pi M(2, 3, 6)$$

$M_i = \overline{m_i}$, so the two lists of the same function name opposite rows

sum of products

group the 1s

a \ bc	00	01	11	10
0	1	1	0	0
1	1	1	1	0

$$f = \bar{b} + ac$$

product of sums

group the 0s

a \ bc	00	01	11	10
0	1	1	0	0
1	1	1	1	0

$$f = (a + \bar{b})(\bar{b} + c)$$

size $1, 2, 4, 8, \ldots$

1	1	0	0
1	1	0	0

as large and as few as possible

wrap around

1	0	0	1
1	0	0	1

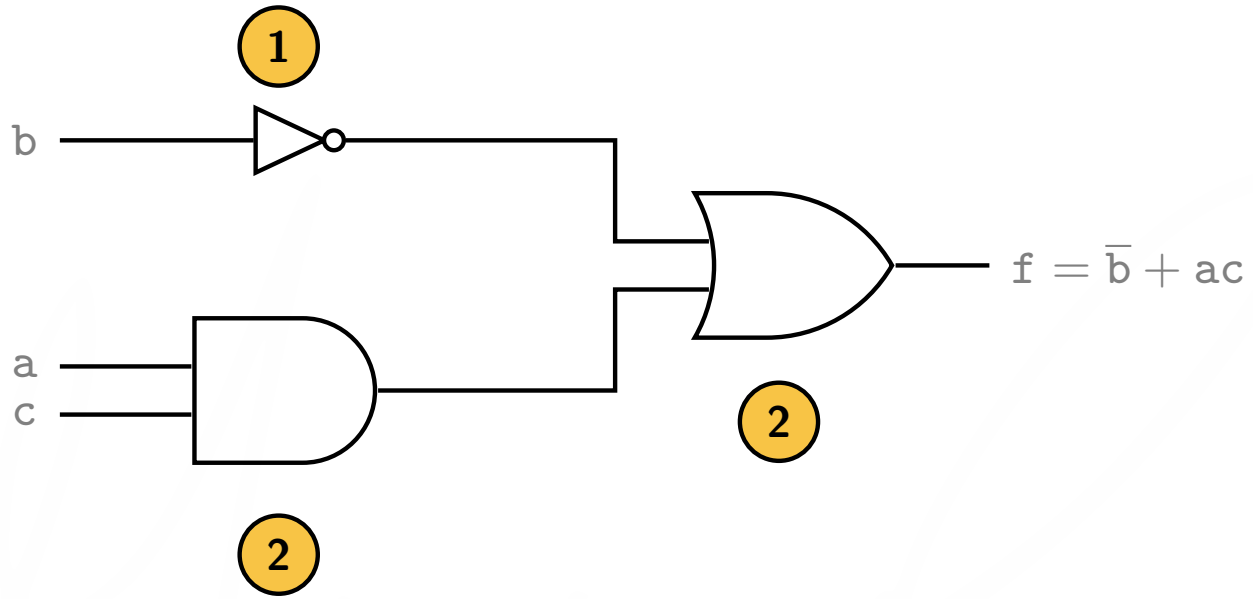
the edges are neighbours

don't-cares

0	1	X	0
0	1	1	0

an X joins a group if it helps

The **cost** of a circuit can be counted in three ways.



the circles count the inputs of each gate

L = 3

literals in the expression

$\bar{b}$, a, c

G = 4

gate inputs, not counting NOTs

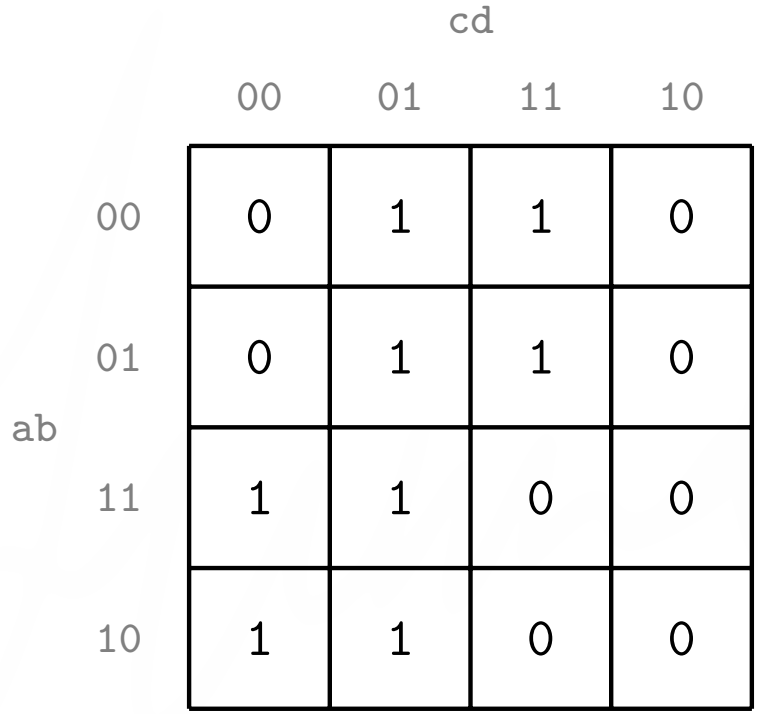
$2 + 2$

GN = 5

gate inputs, NOTs included

$2 + 2 + 1$

question 6



find the minimal sum of products

0 responses

What is the minimal sum of products for this Karnaugh map?

A $\bar{a}d + \bar{c}d$ — 0% — 0 votes

✓ $\bar{a}d + a\bar{c}$ — 0% — 0 votes

C $d + a\bar{c}$ — 0% — 0 votes

D $\bar{a}d + ab\bar{c}$ — 0% — 0 votes

question 7

ab \ cd	00	01	11	10
00	1	0	0	1
01	0	0	0	0
11	0	0	0	0
10	1	1	X	X

find the minimal sum of products

0 responses

What is the minimal sum of products for this Karnaugh map (`X` = don't-care)?

A $\overline{b}\,\overline{d}$ — 0% — 0 votes

B $a\overline{b}\,\overline{c} + \overline{b}\,\overline{d}$ — 0% — 0 votes

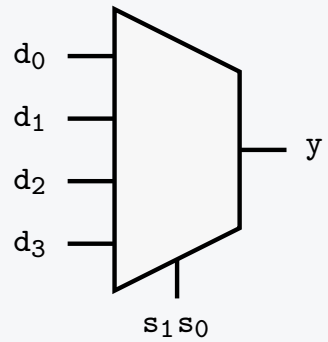
C $a + \overline{b}\,\overline{d}$ — 0% — 0 votes

✓ $a\overline{b} + \overline{b}\,\overline{d}$ — 0% — 0 votes

4

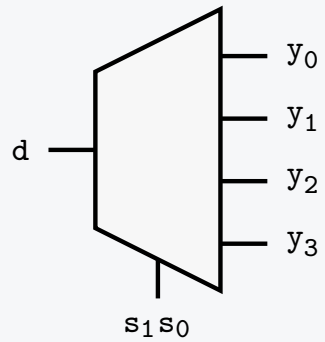
logical devices

multiplexer



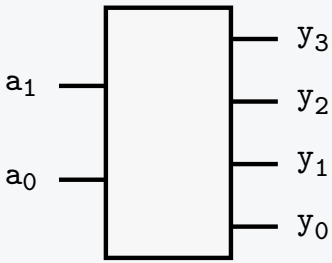
k select bits pick one of 2^k inputs

demultiplexer



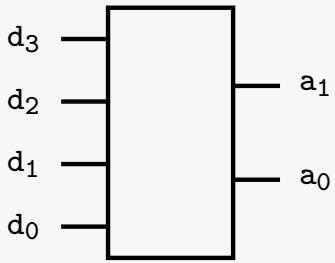
routes d to the selected output

decoder



n bits $\rightarrow$ one of 2^n outputs is 1; $y_i = m_i$

encoder



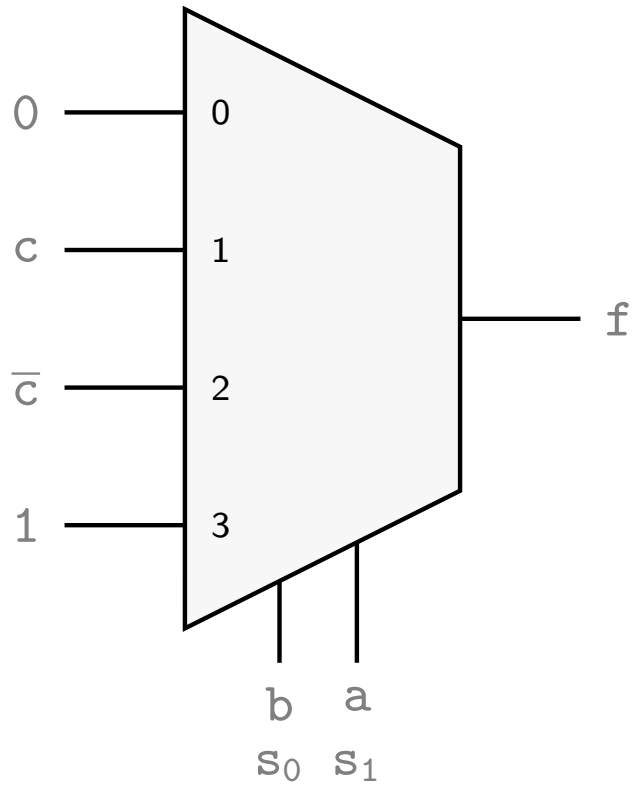
the number of the one input that is 1

0 responses

How many 2-to-1 multiplexers does it take to build a 16-to-1 multiplexer?

A	4	0%	0 votes
B	8	0%	0 votes
✓	15	0%	0 votes
D	16	0%	0 votes

question 9



0 responses

Which function does this multiplexer implement?

A $\Sigma m(2,5,6,7)$ 0% 0 votes

✓ $\Sigma m(3,4,6,7)$ 0% 0 votes

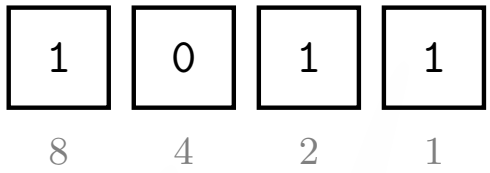
C $\Sigma m(0,3,5,7)$ 0% 0 votes

D $\Sigma m(1,4,6,7)$ 0% 0 votes

5

binary arithmetic

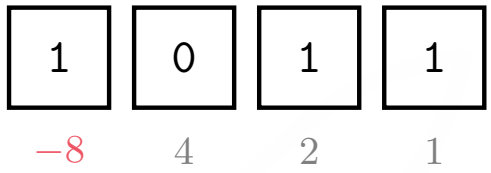
unsigned



$$8 + 2 + 1 = 11$$

range: 0 to $2^n - 1$

signed



$$-8 + 2 + 1 = -5$$

range: -2^{n-1} to $2^{n-1} - 1$

decimal to binary

$11 = 1011$

halve repeatedly; the remainders, read upward, are the bits

binary and hex

1011 0110 $\leftrightarrow$ B6

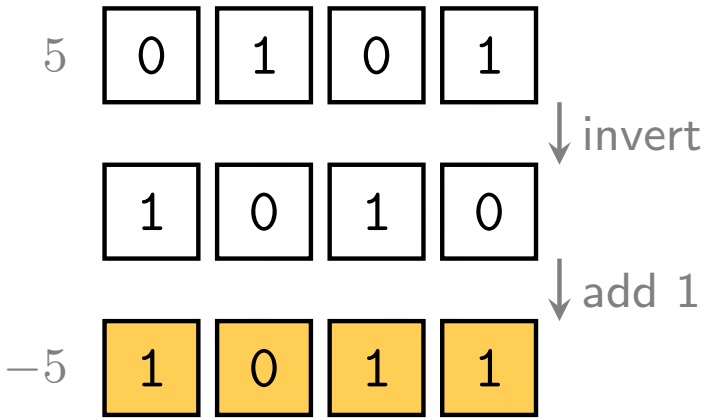
one hex symbol per group of 4 bits; A $= 10, \ldots,$ F $= 15$

binary to decimal

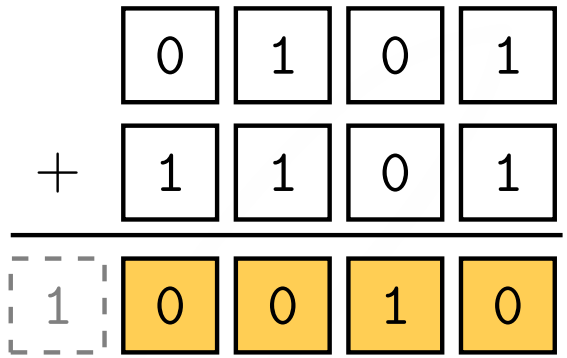
0110 $= 4 + 2 = 6$

add the weights of the 1s; signed, the top weight is negative

negating



adding



$5 + (-3) = 2$; the carry out is dropped

subtracting is adding the negation

$a - b = a + \overline{b} + 1$, the same bit operations for signed and unsigned numbers

0 responses

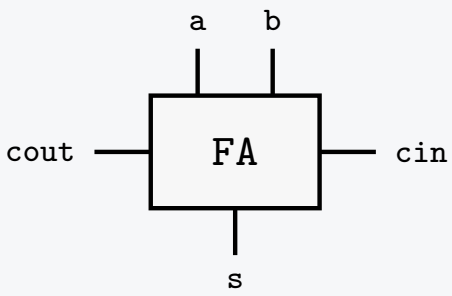
In 5-bit two's complement, what is `00110` − `01011`?

A	`00101`	0%	0 votes
✓	`11011`	0%	0 votes
C	`10101`	0%	0 votes
D	`11010`	0%	0 votes



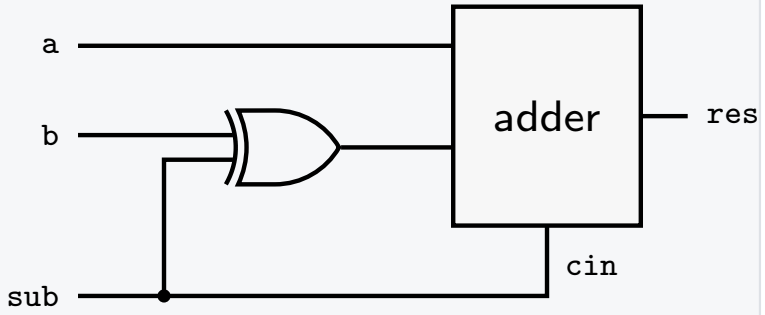
arithmetic circuits

full adder



$$\texttt{s} = \texttt{a} \oplus \texttt{b} \oplus \texttt{cin}$$
$$\texttt{cout} = \texttt{ab} + \texttt{cin}\,(\texttt{a} \oplus \texttt{b})$$

adder/subtractor



$$\texttt{sub} = 1\text{: } \texttt{a} + \overline{\texttt{b}} + 1 = \texttt{a} - \texttt{b}$$

unsigned: read `carry`

$$\begin{array}{r} 1100 \\ +\ 0110 \\ \hline 1\ 0010 \end{array}$$

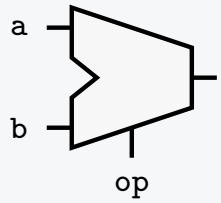
$12 + 6 \neq 2$: a + b is wrong when carry $= 1$
(a − b is wrong when carry $= 0$)

signed: read `overflow`

$$\begin{array}{r} 0110 \\ +\ 0011 \\ \hline 1001 \end{array}$$

$6 + 3 \neq -7$: two numbers of the same sign gave a result of the other sign

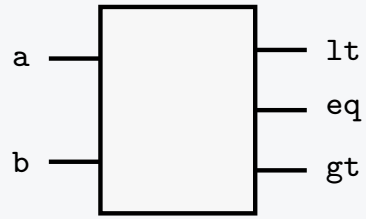
ALU



	useb	inv	cin
a	0	0	0
a + 1	0	0	1
a − 1	0	1	0
a + b	1	0	0
a − b	1	1	1

plus bitwise AND, OR, NOT;
a mux picks the result

comparator



1011 vs 1101: lt

the first bit from the left where they differ decides

shifter

1101 << 1 1010 $\times 2$

1101 >> 1 0110 logical

1101 >> 1 1110 arithmetic

multiplier

$$\begin{array}{r} 101 \\ \times\ 011 \\ \hline 101 \\ 1010 \\ \hline 1111 \end{array}$$

one shifted row per bit of b; the product needs $2n$ bits

signed: sign-extend, subtract the last row

0 responses

Which 4-bit **signed** addition overflows?

A `1100` + `1110` 0% 0 votes

B `0101` + `1101` 0% 0 votes

✓ `0110` + `0011` 0% 0 votes

D `0111` + `1001` 0% 0 votes

0 responses

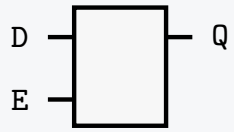
The 4-bit adder/subtractor computes `1000` − `0001`. What are the `carry` and `overflow` flags?

A	carry 0, overflow 0	0%	0 votes
B	carry 1, overflow 0	0%	0 votes
C	carry 0, overflow 1	0%	0 votes
✓	carry 1, overflow 1	0%	0 votes

7

latches and flip-flops

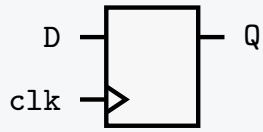
D latch



E	Q_t
0	Q_{t-1}
1	D

follows D while $E = 1$

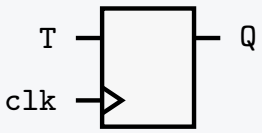
D flip-flop



D	Q_t
0	0
1	1

copies D on the clock edge

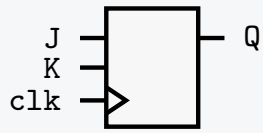
T flip-flop



T	Q_t
0	Q_{t-1}
1	$\overline{Q}_{t-1}$

toggles when $T = 1$: $D = T \oplus Q$

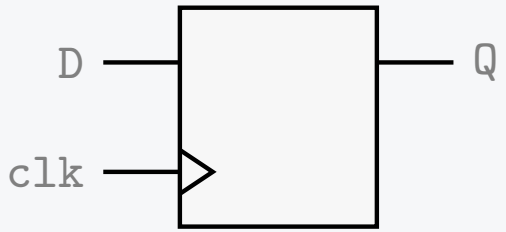
JK flip-flop



J	K	Q_t	
0	0	Q_{t-1}	hold
0	1	0	reset
1	0	1	set
1	1	$\overline{Q}_{t-1}$	toggle

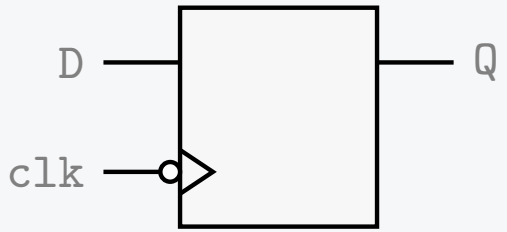
$D = J\overline{Q} + \overline{K}Q$

positive edge



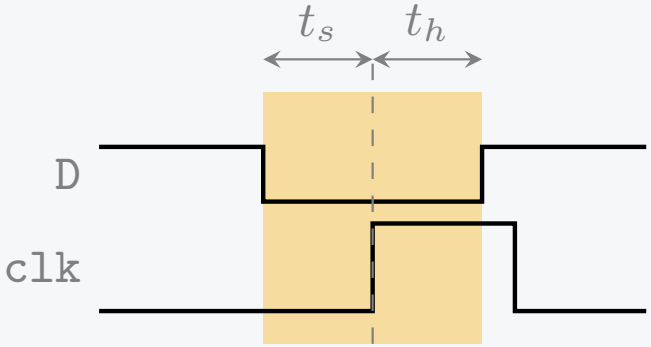
acts as clk rises

negative edge



acts as clk falls (the bubble)

setup and hold times

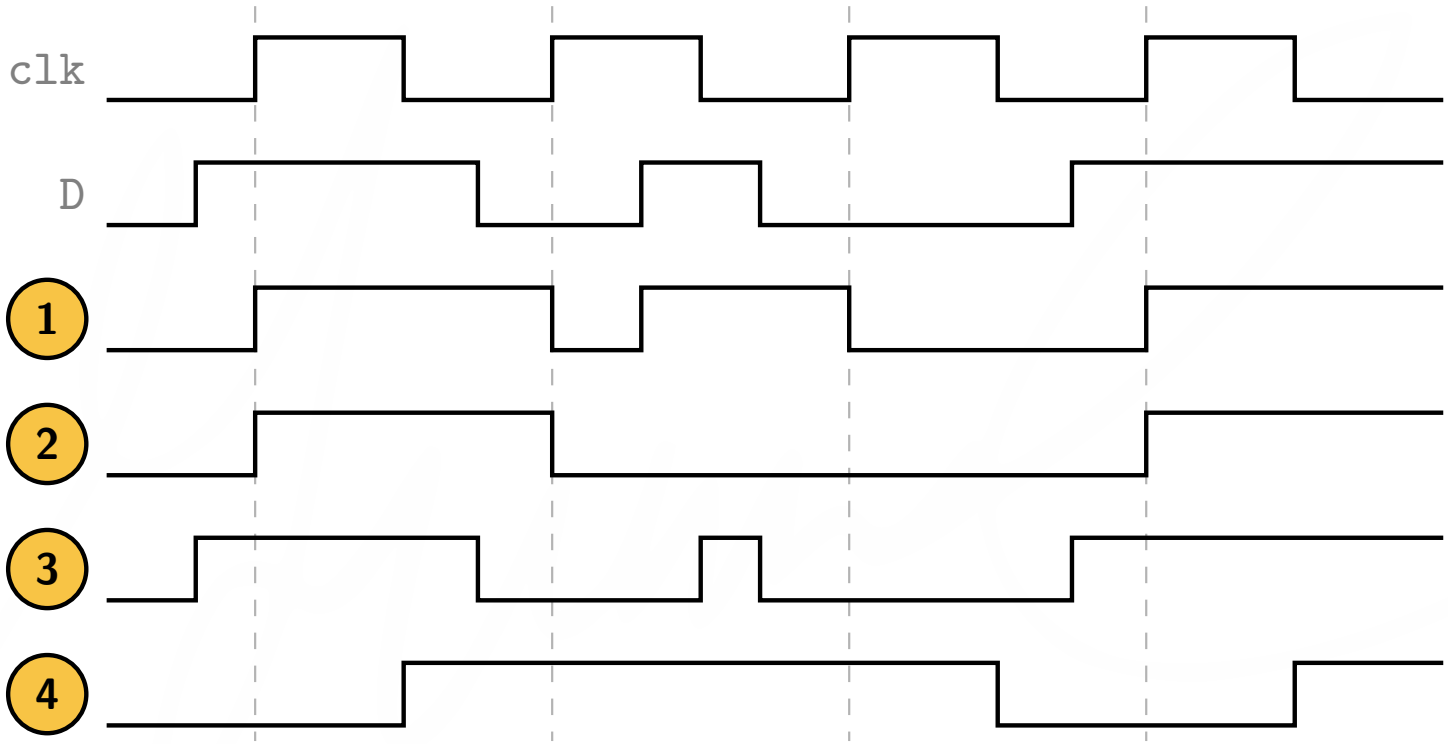


setup time t_s: D is stable this long *before* the edge

hold time t_h: D stays stable this long *after* the edge

if D changes in the shaded window, the stored value is unreliable

question 13



`Q` starts at `0`; which waveform is `Q`?

0 responses

Which waveform is `Q` for a **positive-edge** D flip-flop?

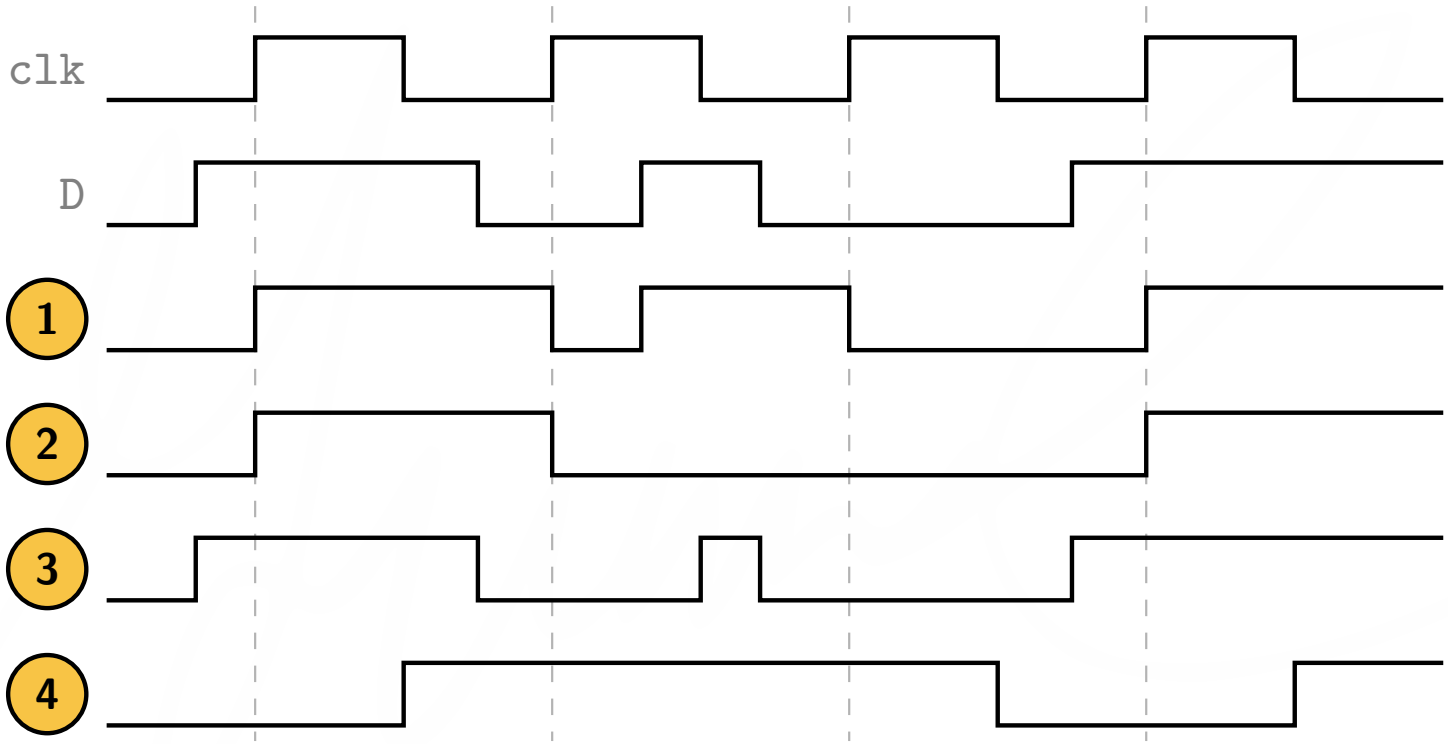
A	1	0%	0 votes
✓	2	0%	0 votes
C	3	0%	0 votes
D	4	0%	0 votes

0 responses

A JK flip-flop starts at `Q` = 0. On six rising edges, (J, K) is 10, 00, 11, 01, 11, 11. What is `Q` after each edge?

A	`110011`	0% 0 votes	B	`101010`	0% 0 votes
✓	`110010`	0% 0 votes	D	`111010`	0% 0 votes

question 15



`Q` starts at `0`; which waveform is `Q`?

0 responses

Same waveforms: which one is `Q` for a **negative-edge** D flip-flop?

A 1	B 2	C 3	✓ 4
0%	0%	0%	0%
0 votes	0 votes	0 votes	0 votes

0 responses

A circuit's clock period is 10 ns; its flip-flops have a setup time of 2 ns and a hold time of 1 ns. The logic feeding `D` settles 9 ns after each rising edge. What happens?

A it works 0% 0 votes

✓ setup-time violation 0% 0 votes

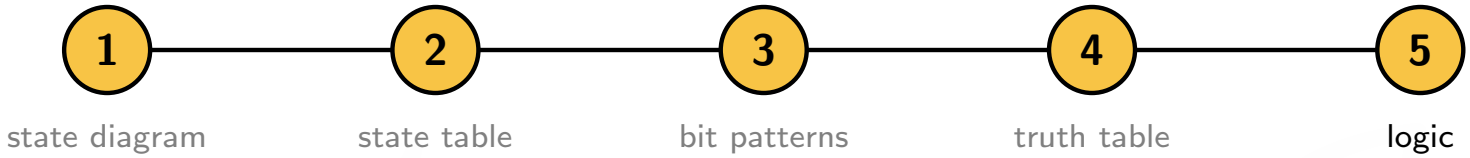
C hold-time violation 0% 0 votes

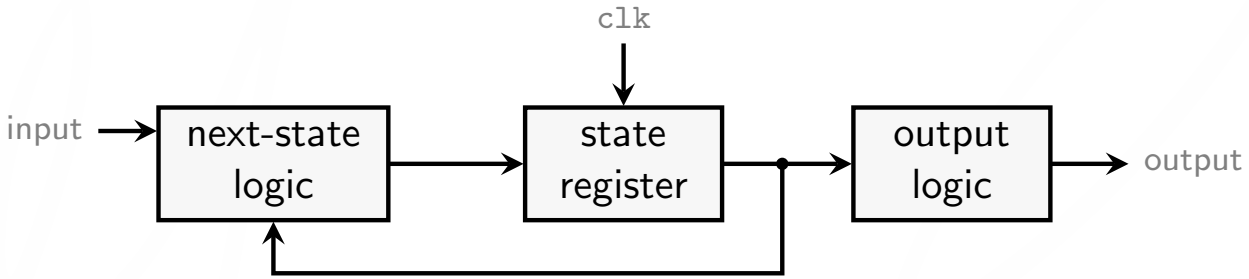
D both violations 0% 0 votes

8

finite state machines

designing an FSM



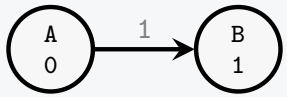


flip-flops needed

$\lceil \log_2 k \rceil$

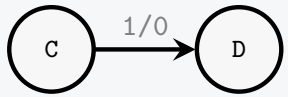
for k states (5 to 8 states need 3)

Moore



output depends on the state only

Mealy



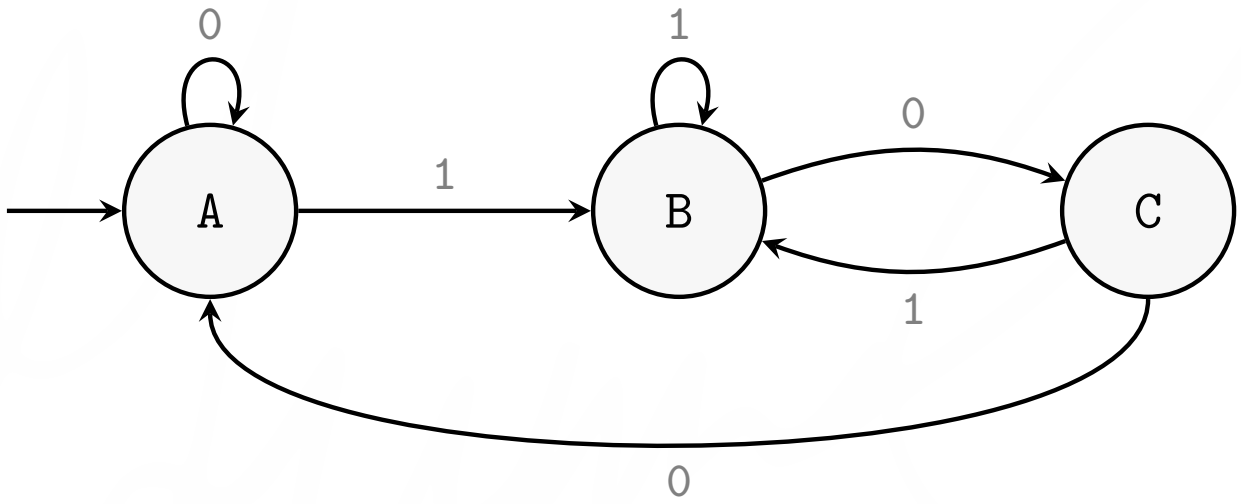
output depends on the state and input

0 responses

A machine has 5 states and one input. How many rows does its truth table (step 4 of the design procedure) have?

A	5	0%	0 votes
B	10	0%	0 votes
✓	16	0%	0 votes
D	32	0%	0 votes

question 18



the machine starts in A; the input is 1, 0, 1, 1, 0, 1, 0, 0, 1

0 responses

The machine starts in A and reads `1`, `0`, `1`, `1`, `0`, `1`, `0`, `0`, `1`. After how many of the nine inputs is it in state C?

A 1	B 2	✓ 3	D 4
0%	0%	0%	0%
0 votes	0 votes	0 votes	0 votes

question 19

Q_1	Q_0	x	D_1	D_0
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0
1	0	0	1	0
1	0	1	1	1
1	1	0	1	1
1	1	1	0	0

a 2-bit counter: it counts up when $x = 1$ and holds when $x = 0$

which equation gives D_1?

0 responses

Which equation gives D_1?

A $Q_1 \oplus x$ — 0% — 0 votes

B $Q_0 x$ — 0% — 0 votes

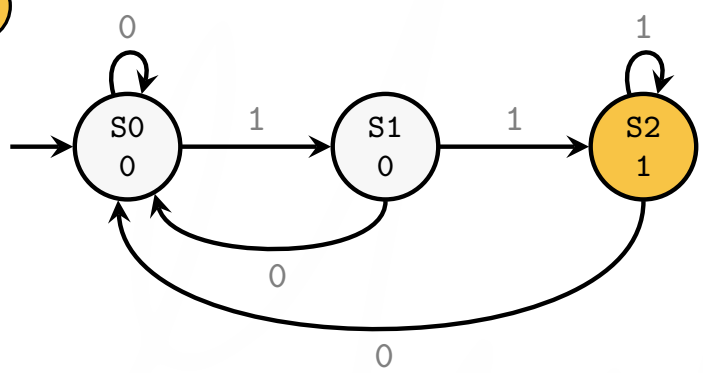
✓ $Q_1 \oplus Q_0 x$ — 0% — 0 votes

D $Q_1 + Q_0 x$ — 0% — 0 votes

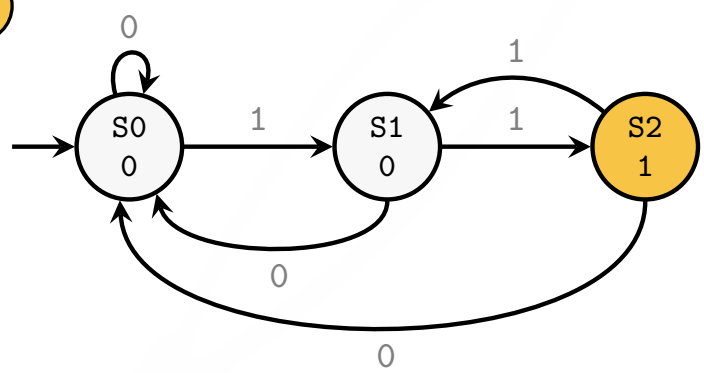
question 20

output 1 exactly when the last two inputs were both 1

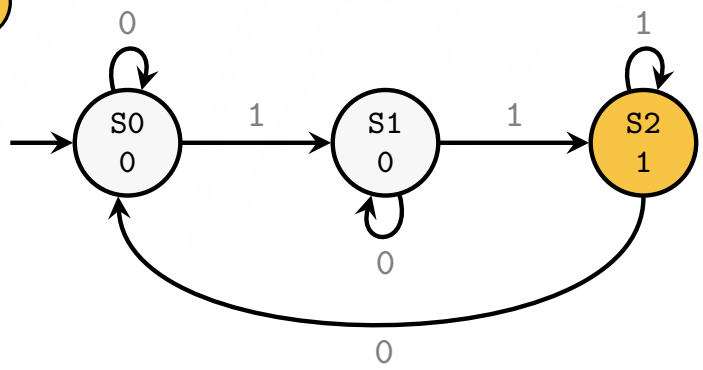
1



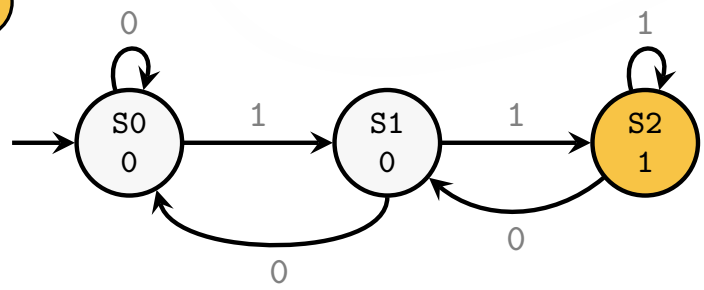
2



3



4



0 responses

Which diagram outputs `1` exactly when the last two inputs were both `1`?

✓ 1	B 2	C 3	D 4
0%	0%	0%	0%
0 votes	0 votes	0 votes	0 votes

putting it all together

0 responses

A 2-to-4 decoder takes inputs `a` (high bit) and `b`. Its outputs y_1 and y_2 feed an `OR` gate. Which single gate is this circuit?

A	AND	0%	0 votes
B	XNOR	0%	0 votes
✓	XOR	0%	0 votes
D	NAND	0%	0 votes

0 responses

On a 4-bit ALU, `a` = `0011` and `b` = `1010` are signed. What does `a − b` give, and does it overflow?

A `1001`, no overflow — 0% (0 votes)

B `0111`, no overflow — 0% (0 votes)

✓ `1001`, overflow — 0% (0 votes)

D `1101`, overflow — 0% (0 votes)

0 responses

Two rising-edge T flip-flops start at Q_1Q_0 = `00`, with $T_0 = 1$ and $T_1 = Q_0$. What is Q_1Q_0 after three clock edges?

A	01	0%	0 votes
B	10	0%	0 votes
✓	11	0%	0 votes
D	00	0%	0 votes

0 responses

A Moore machine outputs `1` after two `1`s in a row. Its states are S0 = `00`, S1 = `01`, S2 = `10` (bits Q_1Q_0). On $x = 1$ it moves S0 → S1 → S2 and stays in S2; on $x = 0$ it returns to S0. Using D flip-flops, what is D_1?

A xQ_0 0%
0 votes

B $Q_1 + x$ 0%
0 votes

✓ $x(Q_1 + Q_0)$ 0%
0 votes

D $x\overline{Q_1}$ 0%
0 votes

0 responses

Simplify $\overline{\overline{a} + \overline{b}} + \overline{a}b$. What is the result, and its gate-input cost G?

A	$a + b$, G = 2	0%	0 votes
B	$ab + b$, G = 4	0%	0 votes
C	b, G = 2	0%	0 votes
✓	b, G = 0	0%	0 votes

0 responses

A 4-bit subtractor computes $a - b$ on **signed** numbers, and its sign bit is used as the test $a < b$. For which pair does this test give the wrong answer?

A `a = 0011, b = 0101` 0% 0 votes

✓ `a = 0111, b = 1000` 0% 0 votes

C `a = 1110, b = 0010` 0% 0 votes

D `a = 0101, b = 0101` 0% 0 votes

0 responses

Two 4-bit signed numbers, `1101` and `0011`, are multiplied into an 8-bit product. What is it?

A	`00100111`	0% 0 votes	B	`11111001`	0% 0 votes
✓	`11110111`	0% 0 votes	D	`10001001`	0% 0 votes

0 responses

For $f(a, b, c) = \Pi M(0, 1, 2, 3, 5, 7)$, what is the minimal product of sums?

A $a + c$ 0% 0 votes

$a\,\overline{c}$ 0% 0 votes

C $\overline{a} + c$ 0% 0 votes

D $a\,(b + \overline{c})$ 0% 0 votes

0 responses

A JK flip-flop with J = K = 1 is clocked at 100 MHz. What is the frequency of its output `Q`?

A	200 MHz	0%	0 votes
B	100 MHz	0%	0 votes
✓	50 MHz	0%	0 votes
D	25 MHz	0%	0 votes

0 responses

A Mealy machine and a Moore machine each output `1` whenever the last three inputs were `1`, `0`, `1` (overlaps allowed). What are the fewest states each needs (Mealy, Moore)?

A	(2, 3)	0% 0 votes	
B	(3, 3)	0% 0 votes	
✓	(3, 4)	0% 0 votes	
D	(4, 4)	0% 0 votes	

Good luck on the **midterm**!